Practice workbook · Using vector ratios to compare areas

Using length ratios found with vectors, with half ab sin C or shared heights, to compare areas of triangles and of the whole figure.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-08

1. Question 10 · part c · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part c only.

10

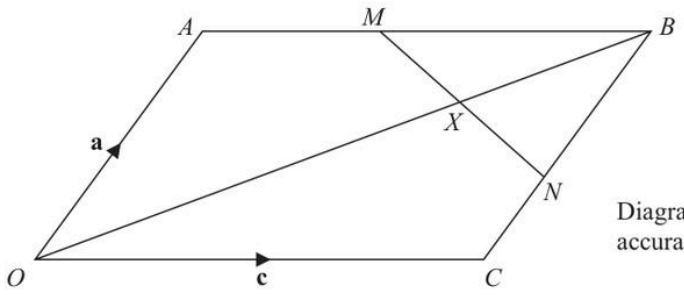


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the parallelogram $OABC$

$$\overrightarrow{OA} = \mathbf{a} \qquad \overrightarrow{OC} = \mathbf{c}$$

The midpoint of AB is M and the midpoint of BC is N.

The line OB intersects MN at the point X.

(a) Find in terms of **a** and **c**,

(i) $\overrightarrow{OB}$

(ii) $\overrightarrow{MN}$

(2)

Given $\overrightarrow{MX} = \lambda\overrightarrow{MN}$ and that $\overrightarrow{OX} = \mu\overrightarrow{OB}$,

(b) use a vector method to find the value of λ and the value of μ.

(8)

(c) Hence find, in its simplest form, the ratio

Area of quadrilateral $OXNC$: Area of parallelogram $OABC$.

(3)

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1. Question 10 · part c · 3 marks

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Worked-example retry · Attempt part c only.

Question 10 continued

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1. Question 10 · part c · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part c only.

Question 10 continued

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1. Question 10 · part c · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 13 marks)

1. Question 10 · part c · 3 marks · Working and answer

Working and answer

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2. Question 11 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

11

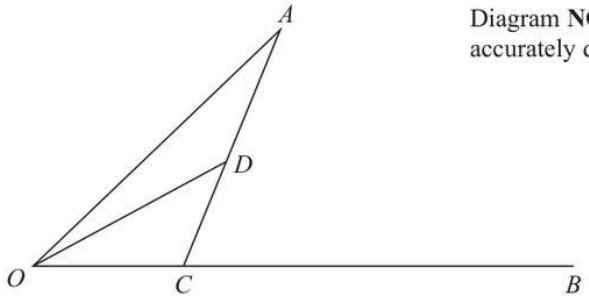


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

The point C divides OB in the ratio $1 : 3$
The point D is the midpoint of AC

(a) Find, as a simplified expression in terms of **a** and **b**

(i) $\overrightarrow{AC}$ (ii) $\overrightarrow{OD}$ (iii) $\overrightarrow{BD}$

(5)

The point E is such that $\overrightarrow{OE} = \lambda\overrightarrow{OA}$

Given that E, D and B are collinear

(b) find the value of λ

(4)

Given that $\dfrac{\text{area } \Delta OAC}{\text{area } \Delta OEB} = \mu$

(c) find the value of μ

(4)

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2. Question 11 · part c · 4 marks

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Further practice · Attempt part c only.

Question 11 continued

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2. Question 11 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 13 marks)

2. Question 11 · part c · 4 marks · Working and answer

Working and answer

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3. Question 8 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

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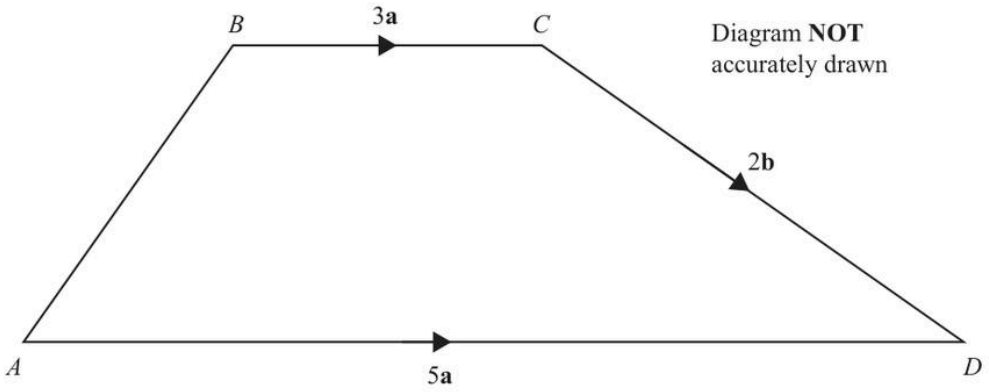


Figure 3

Figure 3 shows a trapezium $ABCD$

$$\overrightarrow{BC} = 3\mathbf{a} \qquad \overrightarrow{AD} = 5\mathbf{a} \qquad \overrightarrow{CD} = 2\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**

(1)

The diagonals BD and AC intersect at point X where $\overrightarrow{BX} = k\overrightarrow{BD}$

(b) Using a vector method, find the value of k

(5)

(c) Find the ratio of the area of triangle CXD : area of the trapezium $ABCD$

(4)

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3. Question 8 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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3. Question 8 · part c · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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3. Question 8 · part c · 4 marks

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Further practice · Attempt part c only.

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Question 8 continued

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(Total for Question 8 is 10 marks)

3. Question 8 · part c · 4 marks · Working and answer

Working and answer

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4. Question 8 · part c · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

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Diagram **NOT** accurately drawn

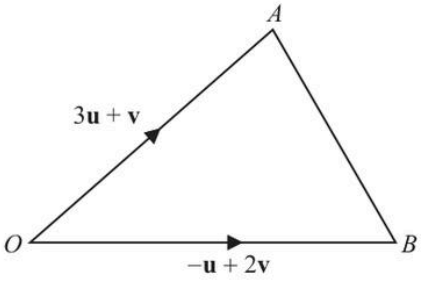


Figure 2

Figure 2 shows triangle OAB with

$$\overrightarrow{OA} = 3\mathbf{u} + \mathbf{v} \qquad \overrightarrow{OB} = -\mathbf{u} + 2\mathbf{v}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ (2)

Point C is such that AC is parallel to OB

Given that $\overrightarrow{OC} = \mu\mathbf{v}$

(b) find the value of μ (4)

The lines OC and AB intersect at point X

Point D lies on $\overrightarrow{OC}$ such that

area of triangle BOX : area of triangle $BXD = 2 : 3$

Using a vector method,

(c) find $\overrightarrow{BD}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ (5)

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4. Question 8 · part c · 5 marks

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Further practice · Attempt part c only.

Question 8 continued

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4. Question 8 · part c · 5 marks

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Further practice · Attempt part c only.

Question 8 continued

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4. Question 8 · part c · 5 marks

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Further practice · Attempt part c only.

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Question 8 continued

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(Total for Question 8 is 11 marks)

4. Question 8 · part c · 5 marks · Working and answer

Working and answer

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5. Question 10 · part c · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

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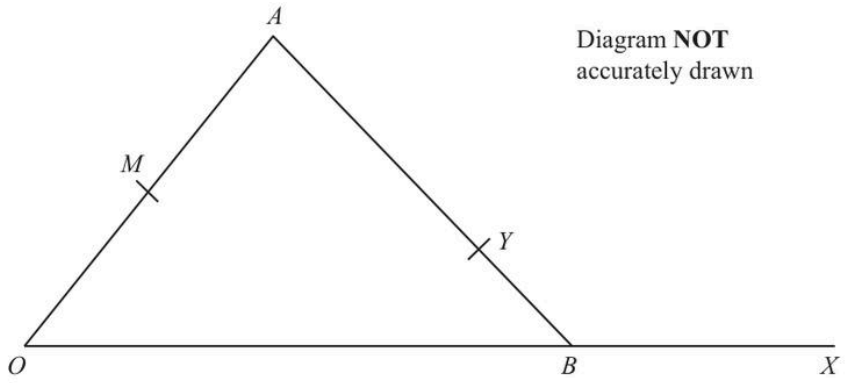


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = 2\mathbf{a}$, $\overrightarrow{OB} = 4\mathbf{b}$ and M is the midpoint of OA.

The point Y lies on AB such that $AY : YB = 3 : 1$

The point X lies on OB produced.

(a) Find as simplified expressions in terms of **a** and **b**

(i) $\overrightarrow{AB}$ (ii) $\overrightarrow{MY}$

(3)

The points M, Y and X are collinear.

(b) Find the ratio $OB : OX$

(5)

(c) Find the ratio of $(\text{Area } \Delta YBX) : (\text{Area } \Delta OAX)$

(3)

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5. Question 10 · part c · 3 marks

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Further practice · Attempt part c only.

Question 10 continued

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5. Question 10 · part c · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 11 marks)

5. Question 10 · part c · 3 marks · Working and answer

Working and answer

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6. Question 10 · part c · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

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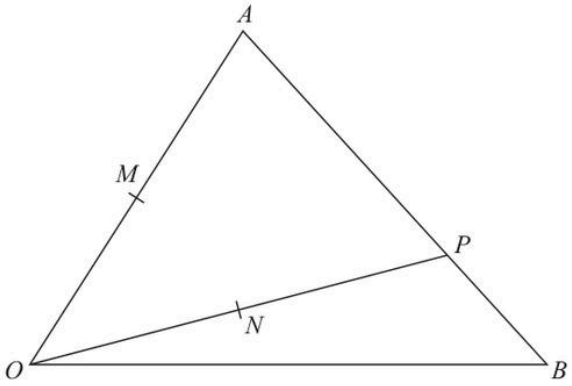


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows triangle OAB in which

$$\overrightarrow{OA} = \mathbf{a} \text{ and } \overrightarrow{OB} = \mathbf{b}$$

The point P lies on AB such that $AP : PB = 3 : 1$

The point M is the midpoint of OA and the point N is the midpoint of OP.

(a) Find, as simplified expressions in terms of **a** and **b**, the vector

(i) $\overrightarrow{OP}$ (ii) $\overrightarrow{MN}$

(4)

The point C lies on OB such that ANC is a straight line.

(b) Using a vector method, find the vector $\overrightarrow{OC}$ as a simplified expression in terms of **b**

(6)

Given that $\dfrac{\text{area of quadrilateral } AMNP}{\text{area of triangle } OAB} = K$

(c) find the exact value of K

(4)

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6. Question 10 · part c · 4 marks

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Further practice · Attempt part c only.

Question 10 continued

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6. Question 10 · part c · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 14 marks)

6. Question 10 · part c · 4 marks · Working and answer

Working and answer

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7. Question 9 · part c.ii · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c.ii only.

9 The points A, B, C and D are the vertices of a quadrilateral where

$$\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b} \qquad \overrightarrow{AC} = 7\mathbf{a} + 7\mathbf{b} \qquad \overrightarrow{AD} = 10\mathbf{a} + 6\mathbf{b}$$

(a) Show that $ABCD$ is a trapezium.

(4)

Given that $|\mathbf{a}| = 1$ and $|\mathbf{b}| = 1$ and that $\mathbf{a}$ and $\mathbf{b}$ are perpendicular to each other

(b) find a unit vector parallel to $\overrightarrow{BD}$

(4)

The point Y lies on CD such that $CY : YD = 1 : 2$

The lines AC and BY intersect at the point X

(c) (i) Use a vector method to find the ratio $BX : XY$

(4)

(ii) Hence find, in its simplest form, the ratio

area of triangle CXY : area of trapezium $ABCD$

(4)

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7. Question 9 · part c.ii · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c.ii only.

Question 9 continued

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7. Question 9 · part c.ii · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c.ii only.

Question 9 continued

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7. Question 9 · part c.ii · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c.ii only.

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Question 9 continued

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(Total for Question 9 is 16 marks)

7. Question 9 · part c.ii · 4 marks · Working and answer

Working and answer

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8. Question 8 · part e · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part e only.

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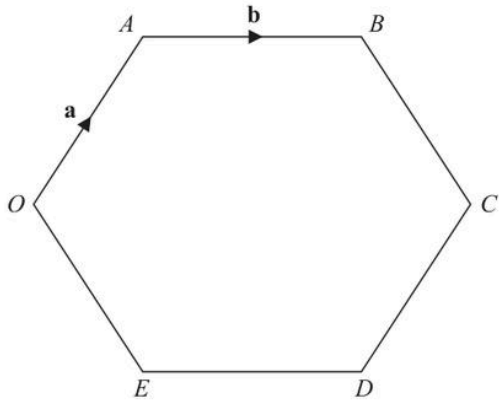


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the regular hexagon $OABCDE$ with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$

(a) Find $\overrightarrow{OB}$ in terms of **a** and **b**

(1)

(b) Find $\overrightarrow{BC}$ as a simplified expression in terms of **a** and **b**

(3)

The point M divides BC in the ratio $2:1$

(c) Find $\overrightarrow{OM}$ as a simplified expression in terms of **a** and **b**

(2)

The point Y is such that OMY and ABY are straight lines.

(d) Use a vector method to find $AB:BY$

(5)

The area of hexagon $OABCDE$ is $60\,\text{cm}^2$

(e) Find the area of triangle OAY

(4)

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8. Question 8 · part e · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part e only.

Question 8 continued

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8. Question 8 · part e · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part e only.

Question 8 continued

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8. Question 8 · part e · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part e only.

Question 8 continued

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(Total for Question 8 is 15 marks)

8. Question 8 · part e · 4 marks · Working and answer

Working and answer

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